

Current

An interactive prototype of a collections and customer-communication platform for a UK energy supplier – what it is, what it is arguing, and how to walk a client through it in ten minutes.

OPEN IT

mosesdex.github.io/current-arrears-prototype

SOURCE

github.com/mosesdex/current-arrears-prototype

SURFACES

Agent workspace · Admin console · Customer app

DATA

Entirely invented

01 What this is

A supplier contacts customers in arrears by text through a third-party conversational-messaging platform. That platform went down for two to four weeks. The immediate problem was that outbound collections messaging stopped. **The more serious problem is what the outage exposed about the arrangement underneath it.**

The instinct after a vendor outage is to buy a more capable vendor. This prototype argues the opposite: that the layer which failed was not the carrier network – the UK carrier layer is served by more than thirty tier-one aggregators and is already redundant – but the application sitting on top of it. The vendor *was* the workflow, so there was nothing to fail over to.

So the proposition is not a better messaging supplier. It is owning four things the supplier should never have been renting: **the conversation state, the suppression list, the audit trail, and the message templates.** Once those are yours, a provider becomes a routing decision rather than a dependency, and a month-long outage degrades to a change of routing weight.

WHAT THE PROTOTYPE IS NOT

It has no backend, no authentication and no persistence – state lives in the page and resets when you reload it. Nothing is connected to a real customer, a real meter or a real provider. It exists to make a set of arguments inspectable and clickable, not to be deployed.

02 The three surfaces

Switch between them from the tabs at the top of the page. Each answers a different question about the same system.

SURFACE	WHOSE SCREEN IT IS	WHAT IT IS THERE TO SHOW
Agent workspace	A collections agent handling live conversations	That the rules protecting customers are enforced in the product rather than written in a policy document — most visibly, that the composer can be blocked by policy rather than by permission.
Admin console	An operations manager	The configuration, oversight and provider-resilience layer: workflow gates, message routing and failover, channels, roles, imports, and the audit trail.
Customer app	The customer in arrears	The authenticated channel that lets every text message stay thin — and, in the annotation column beside the phone, what each screen has to get right.

Two controls in the top rail that change everything

Design notes toggles the annotation layer. On, the prototype explains the rule behind each decision – which is what a client needs. Off, it reads as a plain product, which is what it would be. Show it both ways.

The **moon** icon switches theme. The palette is deliberate and worth a sentence when someone asks about colour: red is the arrears signal, and duty-of-care states are violet *specifically* so that a person needing help is never rendered in the same colour as a debt.

03 The ten-minute demo

If you have one pass through the prototype, use this order. Each step lands one argument, and they build.

1 Open on the agent workspace and say nothing about the product yet

Point at the queue. It is sorted by **how long the customer has been waiting**, not by balance. Sorting by balance makes the people least able to pay wait longest, so the default sort is a policy decision that happens to look like a preference.

Agent workspace · left pane

2 Read the right-hand panel top to bottom on Delyth Morgan

Duty of care sits **above** the money. An agent who reads the balance first has already framed the conversation, so the ordering of that panel is deliberate and is the shortest way to explain what the whole product is trying to be.

Agent workspace · right pane

3 Click Sofia Kaur

She has disputed the balance, and **the composer is gone** – replaced by a statement of why. Not greyed out because this agent lacks a permission: blocked because contesting an amount restricts what may be done with it. Then click Joseph Okafor for a second reason (breathing space) and Account 00483990 for a third (unconfirmed identity, treated as a possible data breach rather than a wrong number).

Agent workspace · three cases in a row

4 Back on Delyth, press Call Delyth — then press Take a payment

The recording **pauses itself** and the red dot goes grey. Card and security numbers must never reach a recording, and the pause is triggered by the payment step opening rather than by the agent remembering. A control that depends on someone remembering is not a control.

Agent workspace · composer · live call bar

5 Tick a couple of the five disclosure steps, press End, choose "Welfare call completed"

The suspension on arrears activity **lifts**, the case advances a stage, and the restriction note in the composer clears. That call is the only thing that could have lifted it – no timer, and no manager override.

Agent workspace · call panel, then the outcome buttons

6 Switch the composer to Email and type a subject naming the balance

Send **disables**. A subject line is a lock-screen preview exactly like an SMS body, so it gets the same minimisation rule. Change it to something neutral and Send comes back.

Agent workspace · composer · Email tab

7 Switch to WhatsApp

Delyth's 24-hour service window closed 31 hours ago, so free text is **disabled** and only a template Meta has already approved can be sent. Scroll the template list to the rejected one: submitted as a utility message, re-categorised by Meta as marketing. The same clause would convert the whole message into direct marketing under UK privacy law. Two unrelated rulebooks, one sentence, the same verdict.

Agent workspace · composer · WhatsApp tab

8 Press Draft with AI

Asked for a payment reminder on this account, **the assistant declines and says why** – and shows the draft it refused to write. The two it does offer are both welfare-call wording, one in Welsh because that preference is on the record. Note the footer: the model was given four case fields, not the customer's name, address or meter number.

Agent workspace · composer · Draft with AI

9 Go to Admin console, then Message routing, and press Advance the scenario five times

Watch the health score fall **while the provider is still returning 200 OK**. That is the failure that hurts and no SLA on the market covers it. The circuit opens, the outbox drains to the second provider paced per recipient, and the conversation survives because it was never stored at the provider.

Admin console · Message routing

10 Finish on the Customer app, Messages screen

One thread across text, app, WhatsApp, email and call. This screen is why every text message can stay thin: the substance lives behind authentication. Read the annotation column beside the phone – it is the client-facing half of the prototype.

Customer app · Messages

IF YOU ONLY HAVE TWO MINUTES

Steps 3, 4 and 6. A composer that blocks itself, a recording that pauses itself, and a subject line that refuses to be sent. All three are the same argument: the rule lives in the product, not in a training deck.

04 The six cases, and what each one is for

The queue is not a list of examples. Each case exists to demonstrate one rule, and together they cover the reasons a collections conversation should stop.

CASE	STATE	THE RULE IT DEMONSTRATES
Delyth Morgan	Care routed	Priority Services Register, medically dependent on electricity, prepayment meter with no top-up for three days. Arrears activity is suspended until a welfare call is completed. Also the only Welsh-language case — Welsh is a Unicode language, so it costs roughly 2.3× per message and that is a floor, not a defect.
Aaron Whitfield	Sending allowed	The ordinary case. A Direct Debit return classified as an affordability signal rather than a dead mandate or our own defect, an arrangement agreed across four channels, and a delivery receipt that says <i>operator</i> rather than <i>handset</i> .
Sofia Kaur	Disputed	Contesting a balance restricts processing to storage and suspends outbound about that balance. Writing to her about the dispute itself is still permitted — and there is an email in the thread doing exactly that.
Joseph Okafor	Breathing space	A statutory moratorium, day 23 of 60, applied at debt level rather than customer level. Contact goes to his authorised adviser. An agent tried to ring him and the dialler refused before it connected.
Account 00483990	Identity unconfirmed	"You have the wrong person." Because mobile numbers get reassigned, this is treated as a possible personal data breach rather than a wrong number, and the channel is suppressed immediately.
Estate of L. Fernandes	Bereavement	A deceased advice code stops every channel instantly with no re-setup prompt — that prompt being the default behaviour of a system that treats a cancelled mandate as a task. The account is also in credit, so it was never eligible for collections at all.

THE POINT OF HAVING SIX

Three of these conversations are open because **the platform is stopping them**, not because an agent is slow. Any response-time target that counted those against a person would push somebody to break the rule that is holding them — which is why the manager view separates "held by policy" from "waiting for an owner".

05 The six channels

A channel is a property of a message, not a separate inbox. Two inboxes for one customer means two versions of what was said, and the one an ombudsman asks for will be the other one.

CHANNEL	EVIDENCE IT PRODUCES	WHAT IT IS HONESTLY FOR
Text	Handset receipt at one provider, operator acknowledgement at the rest	Getting attention, and offering a way back in. Never detail: it lands on a lock screen and possibly a shared device.
App	Read state, behind authentication	Balance, payment, affordability, documents. The only channel that should carry substance.
WhatsApp	Read receipts — the strongest here, though the customer can switch them off	Structured replies via buttons, documents, and customers who will not open anything else. Additive, never load-bearing.
Email	Delivery to a mailbox, which is not reading	Confirmations, adviser correspondence, longer explanations. No payment links, ever.
Call	Strongest — a conversation with a record	Affordability, welfare, anything difficult. The only channel where somebody can say something you did not think to ask about.
Letter	Posting, which is not reading	Statutory notices, and where nothing else is available.

No channel here proves a person read anything. One provider's own documentation admits a switched-off handset still reports delivered, and an email in a spam folder is delivered too – so anything with a statutory deadline keeps a second channel in the evidential path rather than trusting a receipt.

Why WhatsApp is deliberately not load-bearing

WhatsApp has the best reach ratio of the six and is the only one that could be withdrawn by a decision nobody involved is party to. With text, the short code belongs to the supplier, so traffic can move between aggregators. With WhatsApp the provider in front is swappable and **there is no second WhatsApp behind it**. One policy change, one template rejection or one quality downgrade and the channel stops.

Which makes it a near-perfect restatement of the problem this platform was built to solve. So it is built as additive by construction: nothing in the collections process depends on it, and every template that matters also exists as a letter.

06 The admin console, section by section

Operations

Contact to outcome, contact success per channel, and the team table. The funnel is labelled as the wrong shape for this deliberately: **dropping out is not failure** – a customer who reads a message, sees they are fine and does nothing has been served correctly. Contact success is reported per channel because the licence obliges a supplier to review its contact methods where contact has not succeeded.

Team and queues

First-reply times as a **distribution rather than an average**, because an average hides the only cases that matter: thirteen people who waited more than an hour disappear into a mean of six minutes. Plus who is carrying what, what nobody has picked up, and every handover with its reason.

Message routing

Two providers live permanently and a third kept warm. The five-step failover scenario is worth clicking through. Health is scored on **delivery-derived** signals – receipt latency and the terminal-delivered ratio against each provider's own baseline – because every published SLA in this market measures API availability and none measures delivery. One provider's agreement even deems the platform unavailable only during an incident affecting *multiple* customers, so a total outage affecting one supplier alone is contractually not downtime.

Channels

What each channel is honestly for, the call and recording policy, email authentication, bounce classification, and the WhatsApp problem. Note the calls panel: **there is no dialler in this design**. Predictive dialling produces abandoned calls and answer-machine detection produces exactly the silent call the rules exist to prevent. Removing the dialler removes the whole category of breach rather than managing it down to a permitted percentage.

Collections workflow

Five configurable stages and six gates, of which **four cannot be switched off by anyone, including an administrator**. A configurable compliance control is not a compliance control; it is a setting that happens to be correct today.

Message templates

A working linter. Edit the body and the checks re-run: undeclared variables block the send, one promotional clause reclassifies the whole message as direct marketing, pressure language is flagged as both a conduct risk and counter-productive, and the encoding panel shows what a single pasted curly apostrophe costs. Below it, the sender identities that are blocked at operator level – Energy, Bill, Payment, Collection – which no change of provider recovers.

Roles and access

Seven roles against sixteen capabilities. Read the administrator column: it can change a role definition but cannot read a customer conversation without recording a reason, and cannot send a message at all. Separating configuration from operation is what stops the most privileged account being the most dangerous one.

Import

Eight illustrative rows with real failure codes. The landline row is the one to pause on: UK carriers do not bounce SMS to fixed lines because text-to-speech delivery is widespread here, so an arrears message sent to a landline is **read aloud to whoever answers a shared household phone** and returns a delivery receipt. That is an unintended disclosure, not a billing error, which is why line type is screened at import rather than at send.

Contact rules and Audit trail

Three suppression scopes rather than one flag – because a naive STOP handler puts a supplier in breach of its energy licence while it believes it is complying with privacy law. And an append-only hash-chained audit trail, because provider retention runs from twenty-four hours to three months while limitation on a contract debt is six years in England and Wales and five in Scotland, where it extinguishes the obligation outright.

07 What is deliberately absent

The omissions carry as much of the argument as the features. Each of these is a thing a collections platform would normally have.

NOT IN THE PRODUCT	WHY NOT
Agent rankings or scores	Case mix is assigned rather than chosen, so the numbers are not comparable — the agent who takes the hardest conversations looks worst on every throughput measure. A league table also moves behaviour towards the measure rather than towards the customer.
Propensity or likelihood-to-pay scoring	It decides how hard to press a person on the basis of a prediction about them. Nothing in this platform scores a customer, and no automated decision is made about anybody.
A predictive or power dialler	Abandoned calls and answer-machine silence are regulatory exposures traded for agent utilisation, on a call about somebody's arrears.
Keystroke or screen-time monitoring	The research on workplace monitoring finds the effect depends entirely on whether people experience it as support or as surveillance. This product picks support.
Any field that could hold bank details	Not a policy that says we do not store them — the schema has nowhere to put them. A policy can be forgotten; a missing column cannot.
Payment links in email	A collections email carrying a link is indistinguishable from the phishing it invites. The body offers a sign-in and nothing else.
Provider-side conversation storage	The single decision everything else rests on. The thread is keyed to the case, never to a phone number and a sender, so switching provider changes a routing attribute and leaves the conversation, the audit trail and the suppression state untouched.

ONE FEATURE THAT EXISTS SPECIFICALLY TO BE REFUSED

The drafting assistant shows the drafts **it declined to write**, struck through with the reason, rather than hiding them. On a medically dependent account with a self-disconnection risk it will not draft a payment request at all. An assistant that only ever shows you its successes gives you no way to judge it.

08 Scope, limits and provenance

Everything you see is invented

Every customer, balance, meter reference, conversation and phone number is synthetic. Mobile numbers sit inside 07700 900000–900999 and the single landline inside 029 2018 0xxx — the ranges the regulator reserves for fiction — so no fixture can collide with a real subscriber. The photographs a customer appears to have sent are drawn as vector graphics rather than shipped as images, so no real meter or real document is in the file. The synthetic-data badge stays visible in the top rail rather than hiding in a footer.

What it cannot do

- No backend, authentication or persistence — state resets when the page reloads.
- No real provider is connected. The failover scenario is a scripted sequence, not live telemetry.

- The figures in the admin console are illustrative and internally consistent, not measured.
- The drafting assistant's suggestions are written out in advance. No model is called.

Where the reasoning lives

This prototype is the visible half of a larger piece of work. The research behind it – source-cited notes on supplier operations, payment and Direct Debit scheme mechanics, the regulatory position, data protection, provider architecture, agent-workspace evidence and reliability patterns, together with the architecture decision records and a control matrix – sits in a separate private repository. Where that research could not verify something, it says so rather than filling the gap.

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